

Income, Insurance, and Inequality in Oregon:

Insights from the 2020–2024 ACS 5-Year PUMS

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Data and Approach

Data

- ACS PUMS 2020-2024
- n = 198,000
- Oregon Residents

Method

- OSEMN

Tools

- Python: for api call
- R: for EDA and cleaning, survey weights and inflation-adjusted income

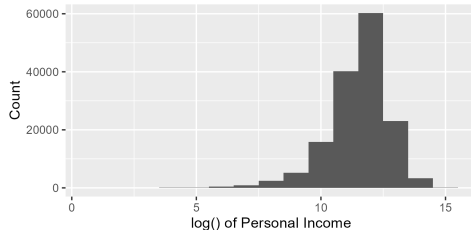
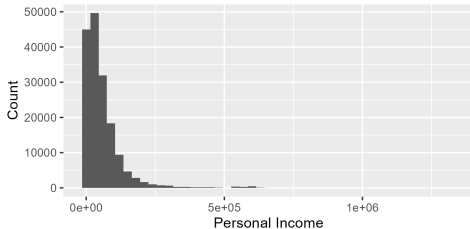
Questions of Interest

Q1: Income predictors

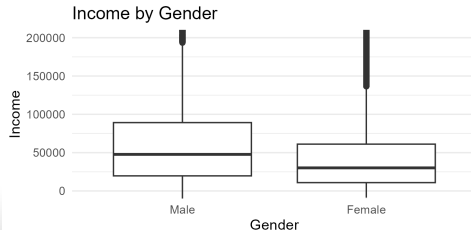
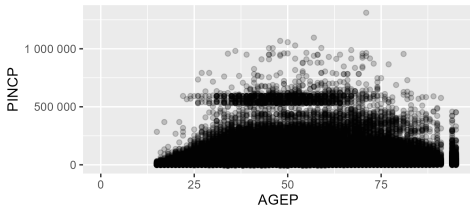
Q2: Insurance coverage gap

Q3: Income vs. transportation

Q1: What predicts income?



Census Data: Personal Income vs Age



Note: Median income: Men ~47K | Women ~30K

Q1: Modeling income

Model equation:

Income = Age + Age² + Education +
Gender + Race + Birthplace +
Household Size

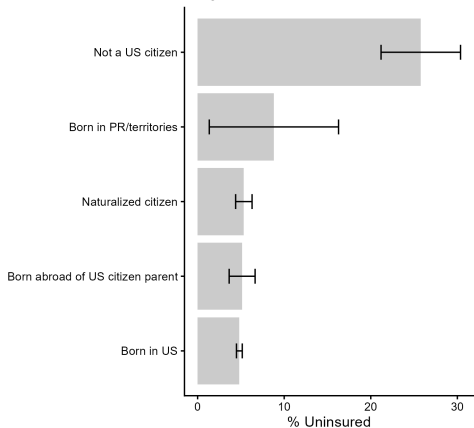
$$R^2 \sim 0.12$$

Key results:

- Education: +2,700 *per year*
- Female: 10,000
- Self-employed: +3,800

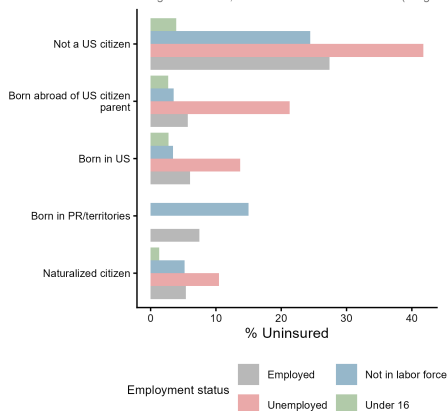
Q2: Who is uninsured in Oregon?

Uninsured Rate by Citizenship Status
Oregon residents, 2020-2024 ACS 5-Year PU



Uninsured Rate by Citizenship and Employm

Oregon residents, 2020-2024 ACS 5-Year PUMS (weighted)



Q2: The gap persists after controls

Full Model: $\text{UNINSURED} \sim \text{CIT} + \text{POVPIP} + \text{SCHL_group} + \text{ESR_group} + \text{NP} + \text{AGEP} + \text{RAC1P} + \text{HISP_binary}$

Reduced Model: $\text{UNINSURED} \sim \text{CIT} + \text{POVPIP}$

Citizenship Status	OR (Reduced)	OR (Full)	Change	p (Full)
Not a US citizen	6.18	4.64	-24.9%	< 0.001
Naturalized citizen	1.16	1.31	+12.9%	0.007
Born in PR/territories	1.92	1.61	-16%	0.339
Born abroad of US citizen parent	1.12	1.14	+1.8%	0.407

Table 2: Model fit comparison: reduced vs full model.

Metric	Reduced Model	Full Model
McFadden R ²	0.075	0.135
Quasi-AIC	81,935	76,687

Q3: Income vs Transportation

Can income be predicted accurately from commute and location information

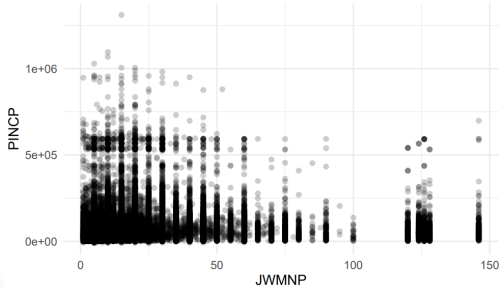


Figure 1: Income vs Commute Time

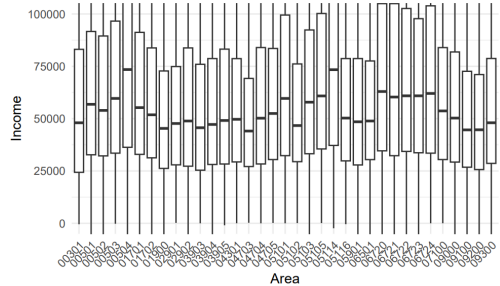


Figure 2: Income by Area Zone

Q3: Modeling

$$PINCP \sim JWMNP + VEH + factor(JWTRNS) + factor(PUMA)$$

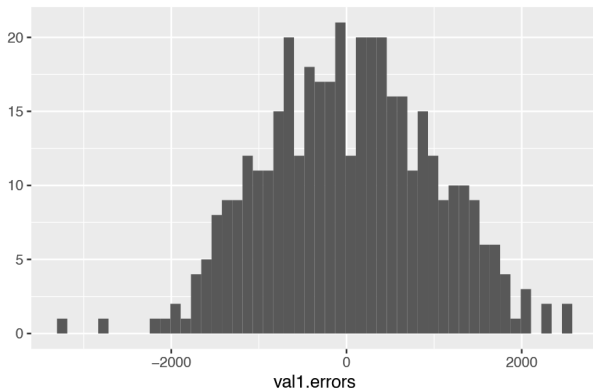


Figure 3: Model Prediction (400 trials)

What we learned

Takeaways:

- Income: Education, age, gender, and labor supply are key predictors
- Insurance: Large, persistent coverage gap for non-citizens
- Transportation: Useful but limited predictor of income

Policy implication: Expanding coverage eligibility could reduce disparities

Data source: U.S. Census Bureau, ACS 2020–2024

Future Directions

Expanding our various models' efficacy by testing on other years, states, census datasets

Obstacles

- Data Ingestion: Programmatic API Calls
- Calculating person weights and inflation adjustment
- Urban vs Rural computation complexity

Thanks!

